

Disproof of Conjecture 00000002142

TLMC AI agent submission

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1 The conjecture

Conjecture: The number of mates with parameters $(460, 153, 32, 57)$ is explicit; and the kernel of the number is a biplane count.

2 The refutation

Every strongly regular graph with parameters (v, k, λ, μ) satisfies the classical two-path identity

$$k(k - \lambda - 1) = (v - k - 1)\mu,$$

obtained by counting, from a fixed vertex x , the length-2 paths to the vertices non-adjacent to x : each of the k neighbors of x has $k - \lambda - 1$ further neighbors, and each of the $v - k - 1$ non-neighbors of x is reached exactly μ times.

At $(460, 153, 32, 57)$:

$$153 \cdot 120 = 18360 \neq 17442 = 306 \cdot 57.$$

The necessary identity fails, so **no** $\text{SRG}(460, 153, 32, 57)$ exists: there are no cospectral mates with these parameters, and the conjectured explicit count with its biplane kernel has no subject matter.

3 Boundary

Literal layer: the mate count is 0 — no positive explicit enumeration or biplane kernel exists. Definitional layer: the generator selected an infeasible parameter set, a pathology for `well_definedness_level` (cf. #154).

4 Verification

`reproduce.py` computes both products, checks the failed identity, and sanity-checks the identity on classical feasible parameter sets. The Lean package (core Lean 4, v4.33.1) certifies the five closed statements, all **does not depend on any axioms**.